Rave Generation

presents




ΩMEGα-50

**User Manual**

# 1. Overview

OMEGA-50 is a six-voice polyphonic synthesizer modeled on the Roland MKS-50, the rack version of the Alpha Juno. Each voice has one oscillator with pulse, sawtooth, sub-oscillator and noise, a high-pass filter, a resonant low-pass filter and an amplifier, shaped by one envelope and one LFO; the voices then pass through a stereo chorus. Its sound was checked against recordings of a real MKS-50.

Our own panel shows every parameter at once, with a live drawing of the envelope at the top.

# 2. Signal Flow

DCO → HPF → VCF → VCA → CHORUS → EQ → DELAY → MULTI-FX → REVERB → VOLUME

The DCO mixes the pulse, sawtooth, sub-oscillator and noise. The HPF and the VCF shape the tone, the VCA sets the level, and the chorus, when it is on, spreads the sound into stereo. The EQ, the delay, the Multi-FX and the reverb on the FX page follow, before VOLUME.

One envelope and one LFO move the pitch, the filter and the level; the knobs in each section set how much.

# 3. The Panel

The sound's controls start at the Default preset, the factory tone A11 PolySynth1; the other controls show their defaults below.

### LFO

| Parameter | Range | Description |
|---|---|---|
| RATE | 0-127 | Speed of the LFO. |
| DELAY | 0-127 | How long after a note starts the LFO fades in. At 0 it is there at once. |

## DCO

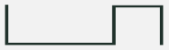
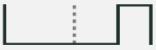
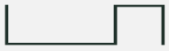
| Parameter | Range | Description |
|---|---|---|
| RANGE | 4', 8', 16', 32' | The octave: 4' is the highest, 32' the lowest. |
| LFO | 0-127 | Vibrato depth. |
| ENV | 0-127 | How far the envelope moves the pitch. |
| ENV MODE | Normal, Inverted, Normal Dyn, Inverted Dyn | Which way the envelope moves the pitch. The Dyn modes scale it by how hard you play. |
| AFTER | 0-127 | Vibrato from aftertouch. |

## KEY

| Parameter | Range | Description |
|---|---|---|
| KEY SHIFT | -12 to +12 | Transposes the sound in semitones. Default 0. |
| DETUNE | -63 to +63 | Fine tune, about -25 to +25 cents. Default 0. |
| MOD SENSE | 0-127 | How much vibrato the mod wheel adds. Default 32. |

## WAVEFORMS

| Parameter | Range | Description |
|---|---|---|
| PULSE | Off, 1-3 | The pulse wave, in the shapes of the chart below. |
| SAW | Off, 1-5 | The sawtooth wave and its variants. |
| SUB | 0-5 | The sub-oscillator's shape. |
| SUB LEVEL | 0-3 | Level of the sub-oscillator. |
| NOISE | 0-3 | Level of the noise. |
| PWM | 0-127 | The pulse width of PULSE 3 and SAW 3, or, with PWM RATE above 0, how far it moves. |
| PWM RATE | 0-127 | Speed of the width modulation. At 0, PWM sets a fixed width. |

| | PULSE | SAW | SUB |
|---|---|---|---|
| 0 | OFF | OFF |  |
| 1 |  |  |  |
| 2 |  |  |  |
| 3 |  |  |  |
| 4 |  |  |  |
| 5 |  |  |  |

The window above PULSE, SAW and SUB shows the shape chosen; click it to pick another, or scroll over it to step through them.

## VOICE

| Parameter | Range | Description |
|---|---|---|
| ASSIGN | Poly, Chord, Mono | Poly plays up to six notes. Chord plays the chord chosen with CHORD from every key. Mono plays one note at a time. Default Poly. |
| CHORD | 1-16 | The chord for Chord mode, listed as semitones from the key you play. 1 is a single note, 15 all six voices in unison. Default 1. |
| PORTA | Off, On | Click PORTA in the VOICE bar; its light shows when it is on. Notes glide from one to the next. Default Off. |
| PORTA TIME | 0-127 | Glide time. Default 20. |

## VCF

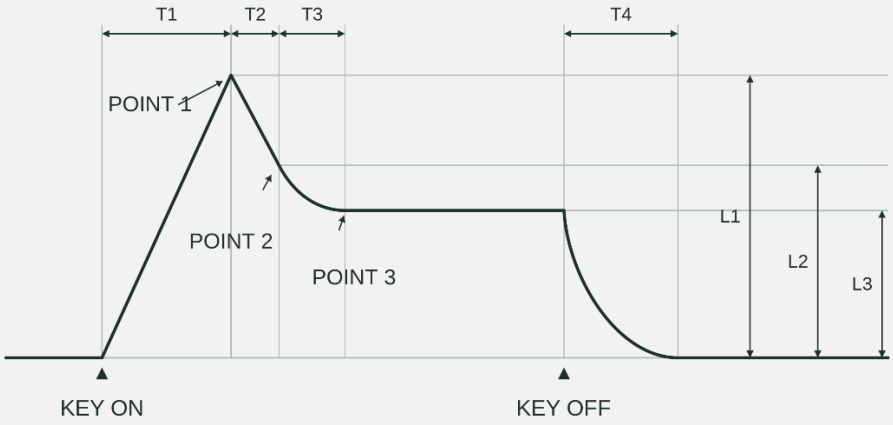
| Parameter | Range | Description |
|---|---|---|
| HPF | 0-3 | High-pass filter: 1 to 3 thin out the low end, more at each step; 0 adds a bass boost. |
| FREQ | 0-127 | Cutoff of the low-pass filter. |
| RES | 0-127 | Resonance, a peak at the cutoff. |
| LFO | 0-127 | How far the LFO moves the cutoff. |
| ENV | 0-127 | How far the envelope moves the cutoff. |
| ENV MODE | Normal, Inverted, Normal Dyn, Dynamics | Which way the envelope moves the cutoff. Normal Dyn scales it by how hard you play; Dynamics moves the cutoff by how hard you play alone, without the envelope. |
| KEY FOLLOW | 0-127 | How far the cutoff follows the keyboard: higher notes open it more. |
| AFTER | 0-127 | Cutoff from aftertouch. |

## VCA

| Parameter | Range | Description |
|---|---|---|
| LEVEL | 0-127 | Voice level. |
| ENV MODE | Env, Gate, Env Dyn, Gate Dyn | Env shapes the level with the envelope; Gate turns it on and off with the key. The Dyn modes add how hard you play. |
| AFTER | 0-127 | Level from aftertouch. |

## ENV

| Parameter | Range | Description |
|---|---|---|
| T1 | 0-127 | Attack time, from zero to L1. |
| L1 | 0-127 | Level at the end of the attack. |
| T2 | 0-127 | Time from L1 to L2. |
| L2 | 0-127 | The second level. |
| T3 | 0-127 | Time from L2 to L3. |
| L3 | 0-127 | Sustain level, held while the key is down. |
| T4 | 0-127 | Release time, from L3 to zero after you let go of the key. |
| KEY FOLLOW | 0-127 | Higher notes run the envelope faster. |




The one envelope serves the pitch (DCO ENV), the filter (VCF ENV) and the level (VCA in its Env modes).

## ENVELOPE DISPLAY

The display at the top of the window draws the envelope as you set it: each T knob sets the width of its part of the curve, each L knob the height of its point. The flat part before KEY OFF stands for the held key, so its length is not a setting. The envelope icon right of the display hides or shows it; the T icon above it turns the labels on and off.

## CHORUS

| Parameter | Range | Description |
|---|---|---|
| CHORUS | Off, On | Click the CHORUS bar; its light shows when the chorus is on. |
| RATE | 0-127 | Chorus speed. |

## BENDER

| Parameter | Range | Description |
|---|---|---|
| RANGE | 0-12 | Pitch-bend range in Poly, in semitones. |
| MONO BEND | 0-12 | Pitch-bend range in Chord and Mono, in semitones. Default 12. |

## OUTPUT

| Parameter | Range | Description |
|---|---|---|
| VOLUME | -40 to +12 dB | Output level. Default 0 dB. |

## FX PAGE

Click FX in the BENDER bar and the bottom row shows the effects page in place of ENV, CHORUS and BENDER; click it again to go back. VOLUME stays where it is. The effects come after the chorus: the EQ, the delay, the Multi-FX, then the reverb. The Multi-FX and the delay are those of our JPFX plugin; each can follow the song tempo with the SYNC menu in its bar. Everything starts neutral, so the sound is unchanged until you turn something. Click a section's name in its bar to switch the section off and on; its knobs dim while it is off.



## EQ

| Parameter | Range | Description |
|---|---|---|
| ON | Off, On | Click EQ in its bar; its light shows when the EQ is on. Switched off, GAIN and the EQ leave the sound as they find it and their knobs keep their settings. Default On. |
| GAIN | 0 to +40 dB | Drives the sound into the distortion of a mixer channel, before the EQ. At 0 dB it is off. Turn VOLUME down to tame the level. Default 0 dB. |
| BASS | 0-127 | Low end of the JPFX tone control; 64 is flat. Default 64. |
| MID FREQ | 220 Hz to 9 kHz | Centre of the mid band; 1.5 kHz at the middle of the knob. Default 1.5 kHz. |
| MID GAIN | -15 to +15 dB | Cuts or boosts the mid band. At 0 dB the band is off. Default 0 dB. |
| SWEEP | 0-100 % | The pill under MID FREQ: how far an LFO moves the mid band while you play. The ring round the knob shows where it goes. You hear it once MID GAIN is away from 0 dB. Default 0 %. |
| RATE | 0.02 to 20 Hz | The pill under MID GAIN: the speed of the sweep. Default 0.5 Hz. |
| TREBLE | 0-127 | Top end of the JPFX tone control; 64 is flat. Default 64. |
| SYNC | Off, 1/16 to 8 Bars | In the EQ bar: the sweep in note values at the song tempo; Off leaves it to RATE. Default Off. |

## MULTI-FX

| Parameter | Range | Description |
|---|---|---|
| ON | Off, On | Click MULTI-FX in its bar; its light shows when the effect is on. Default On. |
| TYPE | 13 effects | Super Chorus Slow, Mid, Fast and Clear; Flanger Slow, Deep and Fast; Deep Phasing Slow; Jet Phasing; Twisting; Freeze Phase 1 and 2; Distortion. Default Super Chorus Slow. |
| LEVEL | 0-127 | How much of the effect; at 0 it is off. Default 0. |
| SYNC | Off, 1/16 to 8 Bars | In the MULTI-FX bar: the sweep of the effect in note values at the song tempo; Off lets it run at its own speed. Default Off. |

## DELAY

| Parameter | Range | Description |
|---|---|---|
| ON | Off, On | Click DELAY in its bar; its light shows when the delay is on. Switched off, the echoes fade out. Default On. |
| TIME | 0-127 | The time between echoes, shown under the knob in ms; the echoes move from left to right. Default 64 (100 ms). |
| FEEDBACK | 0-127 | How many echoes. Default 64. |
| LEVEL | 0-127 | How loud the echoes are; at 0 the delay is off. Default 0. |
| SYNC | Off, 1/16 to 1/2 | In the DELAY bar: the time between echoes in note values at the song tempo; Off leaves it to TIME. Default Off. |

## REVERB

| Parameter | Range | Description |
|---|---|---|
| ON | Off, On | Click the REVERB bar; its light shows when the reverb is on. Switched off, the reverb takes in no more sound and its tail rings out. Default On. |
| PRE-DELAY | 0-250 ms | How long before the reverb sets in. Default 125 ms. |
| DECAY | 0.01-20 s | How long the reverb rings. Default 6 s. |
| MIX | 0-100 % | How much reverb; at 0 % it is off. Default 0 %. |

## Knobs

Hold Shift while dragging a knob for fine adjustment. Right-click a knob to return it to its default (Alt+click works too, Option+click on macOS); double-click it to type a value. Click a selector's value box, or a waveform window, to choose from a list.

## Hints

Hover over a control to read what it does in the bar at the bottom of the window. The (i) icon at the top right turns the hints on and off.

## Menu Bar

Undo and Redo; the panel's opacity, 0% to 100%, to see the DAW behind it; the interface size, 70% to 200%; and the preset menu, which browses, loads and saves presets and copies and pastes them between instances.

# 4. Presets

| Folder | Sounds | Contents |
|---|---|---|
| Default | 1 | The starting sound, factory tone A11 PolySynth1. |
| Factory A | 64 | The original's factory tones, A Group, A11 to A88. |
| Factory B | 64 | B Group, B11 to B88, with the hoover What the at B86. |
| Artists | 47 | 24 by The Viper, 10 by DJ Isaac and 13 by Catscan. |

A preset sets every control. KEY SHIFT, DETUNE, the VOICE settings, MONO BEND, MOD SENSE, VOLUME and the FX page go back to their defaults when you load one.

### Loading .syx Files

Drag a .syx file from your computer onto the envelope display to load the tone in it: a tone dump from an MKS-50 or Alpha Juno, or a 64-tone bank, which loads its first tone. The tone's knobs turn to it and the display shows its name for a moment; KEY SHIFT, DETUNE, the VOICE settings, MONO BEND, MOD SENSE, VOLUME and the FX page stay as they are. Your DAW saves the change with the project, and Undo takes it back. Patch banks and chord memory files hold no tone and are not loaded.

# 5. Playing

OMEGA-50 plays notes on any MIDI channel and answers to:

Velocity: the Dyn modes of the three ENV MODE selectors.

Aftertouch: the three AFTER knobs.

Mod wheel: vibrato, as much as MOD SENSE sets.

Pitch bend: RANGE in Poly, MONO BEND in Chord and Mono.

Sustain pedal: holds the notes.

# 6. Automation

All 67 controls on the panel can be automated in your DAW. The icons at the top right and the FX button cannot.

### Support

For technical support, updates, and additional information:

Website: ravegeneration.io

Email: support@ravegeneration.io

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